PredictStock: AI Demand & Inventory Platform

Comprehensive Technical Architecture & Frontend Implementation Specification

Stack: React 18 • Vite 5 • Recharts • Lucide • Axios • Tailwind / Glassmorphic Engine

1. High-Level Architectural Paradigm

The PredictStock user interface is engineered as an enterprise-grade **Decoupled Single Page Application (SPA)** adhering strictly to **Component-Driven Architecture (CDA)** and **Unidirectional Reactive Data Flow**. The frontend acts as an autonomous visual decision cockpit, running independently from the Python computational backend and communicating strictly over asynchronous JSON REST endpoints.

This decoupling provides three fundamental architectural advantages over monolithic dashboard scripts (such as legacy Streamlit implementations):

- **Zero-Latency Client-Side Manipulation:** Once the active retail universe (13,447 product-store pairs) is ingested into the virtual DOM, multi-dimensional filtering, searching, column sorting, and pagination execute locally in browser memory with instantaneous (< 16ms) response times.
- **Debounced Reactive Recalculation:** Dynamic simulation parameters (e.g., modulating the safety stock buffer from 0% to 100%) are throttled via an 800ms debounce loop, eliminating redundant network overhead while the operator interacts with controls.
- **Stateful Persistence & Dual Theme Engine:** Operator preferences, active scopes, and light/dark theme selections persist locally across browser sessions via localStorage.

> ***CORE ARCHITECTURAL SHIFT:*** *The monolithic Streamlit entry point (app.py) was completely decommissioned. By establishing a dedicated REST interface with an asynchronous FastAPI backend and a modern React 18 single-page application, client rendering is entirely unblocked by Python analytical computations.*

2. Complete Technology Stack & Library Matrix

Layer / Dependency	Version	Technical Function & Architectural Responsibility
React	v18.2.0	Core UI library providing Concurrent Mode, virtual DOM reconciliation, and hooks.
Vite	v5.0.0	Next-gen frontend build tool; native ESM bundling, HMR, and API proxy routing.
Recharts	v2.10.0	Declarative SVG charting library providing responsive charts, donuts, and tooltips.
Lucide React	v0.344.0	Clean SVG icon pack for navigation pills, status badges, and action triggers.
Axios	v1.6.0	Promise-based HTTP client managing API communication, uploads, and binary downloads.
Custom SVG Engine	Native	Hand-crafted vector semicircle arc gauge rendering stock coverage dynamically.
Google Fonts	Web CDN	Inter (primary enterprise typography) and JetBrains Mono (monospaced metrics).

3. Design System & Glassmorphic Styling Philosophy

The interface adopts a high-density, clean enterprise aesthetic inspired by modern mission-critical flight decks and data platforms (Linear, Datadog, PowerBI). Visual hierarchy is maintained through translucent elevation, glowing status indicators, and frosted glass layers.

- **Frosted Glassmorphism:** Elevated surfaces apply backdrop-filter: blur(16px) with semi-transparent base fills (rgba(255, 255, 255, 0.85) in Light Mode and rgba(15, 23, 42, 0.72) in Dark Mode). This preserves context while delivering a modern, layered visual depth.

- **Dual Dynamic Theme Engine:** A comprehensive token system defined in CSS custom properties enables instant switching between Light Theme (optimal for daytime operational review) and Dark Theme (high-contrast cybernetic palette for low-light command monitoring).
- **Tri-Level Status Signaling:** Strict visual conventions govern inventory health: **Critical** (Red #DC2626, < 50% coverage), **Low** (Amber #D97706, 50–100% coverage), and **Sufficient** (Emerald #16A34A, ≥ 100% coverage).

4. Component Hierarchy and Directory Map

Module / Path	Component Name	Role in Presentation Layer
components/	**Navbar.jsx**	Sticky brand banner, live operational status beacon, search bar, theme toggle.
components/	**Sidebar.jsx**	Slide-over optimization drawer: CSV upload, safety buffer slider, store filters.
components/	**KPICard.jsx**	High-impact summary card with colored border accents, icons, and count-ups.
components/	**GlassCard.jsx**	Reusable surface wrapper implementing border glows, shadows, and blur filters.
components/	**StatusBadge.jsx**	Standardized micro-pill indicating Critical, Low, or Sufficient inventory health.
tabs/	**ExecutiveTab.jsx**	Macro-level store analytics, status distribution donuts, and multi-horizon volume.
tabs/	**DeepDiveTab.jsx**	Single-SKU diagnostic cockpit, what-if simulator, arc gauge, and related products.
tabs/	**WorkbenchTab.jsx**	High-density sortable procurement table with pagination, CSV/Parquet exports.
tabs/	**MLBenchmarksTab.jsx**	Untouched test metrics, 3-model benchmark bar chart, and final model performance matrix.
charts/	**StockGaugeChart.jsx**	Mathematical SVG semicircle arc gauge plotting stock coverage with needle.
charts/	**BenchmarkBarChart.jsx**	Grouped multi-horizon benchmark comparing LightGBM, CatBoost, and XGBoost.

5. Operational Cockpit: In-Depth Breakdown of the 4 Tabs

Tab 1: Executive & Store Analytics (Macro Level)

Designed for inventory directors and regional supply chain heads, this view provides cross-network visibility into inventory health and store-level demand balance:

- **Stock Status Distribution (Donut Chart):** Recharts Pie visualization categorizing active pairs into Critical, Low, and Sufficient tiers, displaying both pair counts and percentage shares.
- **Store Branch Comparison (Grouped Bars):** Compares on-hand current stock against required stock and recommended replenishment orders across Stores 1, 2, 3, and 4.
- **Top 12 Urgent Items (Horizontal Bars):** Inverted priority chart ranking the 12 most depleted SKUs requiring immediate cross-dock dispatch.
- **Multi-Horizon Forecast Volume:** Visualizes cumulative customer demand across 7-day, 30-day, and 90-day forward windows.

Tab 2: Product-Store Deep Dive & What-If Simulator

Empowers store managers to inspect individual SKUs, test purchasing scenarios, and analyze substitute products:

- **Clean Store & Product ID Selectors:** Dedicated dropdown filters populated with 13,447 active pairs without cluttering text.
- **Interactive What-If Simulator:** Operators can dynamically modify on-hand stock and adjust safety stock buffers (0–100%) to observe real-time impacts on reorder quantities.
- **Custom SVG Semicircle Arc Gauge:** A standalone vector visualization dividing stock ratios into Danger (< 50%), Caution (50–100%), and Optimal (> 100%) arc segments with an animated needle indicator.

- **Demand Trajectory Curve:** Line chart plotting projected customer absorption across 7, 30, and 90 days with reference lines for simulated on-hand inventory vs. required stock.
- **Category-Matched Related Products Table:** Surfaces alternative SKUs within the same merchandise department to facilitate substitution during severe stockouts.

Tab 3: Replenishment Order Workbench (Procurement Grid)

A high-density operational workbench configured for procurement teams to execute physical replenishment orders:

- **Full-Spectrum Data Grid:** Displays item ID, store ID, department, stock status, on-hand stock, multi-horizon forecasts (7D/30D/90D), safety stock, required stock, and recommended reorder quantities.
- **Multi-Column Sorting & Pagination:** Instant bidirectional sorting across all columns; client-side pagination limited to 50 rows per page for seamless memory performance across 13,447 records.
- **Visual Stock Ratio Bars:** Mini horizontal progress bars indicating coverage ratio with dynamic color shifts (Red $\rightarrow$ Amber $\rightarrow$ Green).
- **One-Click Procurement Exports:** Asynchronous streaming endpoints generate instant **CSV** and **Parquet** procurement order sheets for downstream ERP ingestion.

Tab 4: ML Benchmarks & Model Evaluation

Provides transparency into machine learning validation rigor and model architecture decisions:

- **Untouched Hold-Out Test Highlights:** Elevated KPI metrics summarizing final performance on the 89-day test set (7-Day $R^2 = 0.9346$, 30-Day $R^2 = 0.9292$, Aggregate Volume Bias = -2.26%).
- **Interactive 3-Model Benchmark Chart:** Grouped bar chart comparing LightGBM, CatBoost, and XGBoost with interactive pill toggles for R^2 Score, MAE, and RMSE.
- **Final Model Performance & Specs Matrix:** High-density production table detailing 7D, 30D, and 90D test R^2, MAE, RMSE, and volume bias alongside architecture, feature space, and inference speed specifications.

6. State Management, Ingestion Pipeline & Event Lifecycle

State management in App.jsx coordinates data synchronization, client-side caching, and asynchronous server interaction:

1. **Initial Mount & Ingestion:** Upon application load, a single HTTP call to POST /api/compute retrieves the baseline 13,447 forecast pairs and precomputed KPI summaries.
2. **Instantaneous Client-Side Filtering:** A memoized hook (useMemo) filters the raw row array against active store selections, status checkboxes, and search queries without triggering server queries.
3. **Debounced Policy Re-computation:** When the operator adjusts the global safety stock buffer, an 800ms debounce timer waits for input stabilization before dispatching a re-computation request to the backend.
4. **Live Stock CSV Upload:** Operators can upload live store inventory CSV files; the backend validates schema integrity, recalculates net shortfalls, and updates the reactive frontend state in real time.

7. Production Build Verification & Performance Metrics

Build Parameter / Asset	Observed Production Output	Engineering Assessment
Bundler & Build Tool	Vite v5.4.21 (Production Build)	Rollup-driven code splitting and minification.
Transformed Modules	2,341 React & SVG modules	Zero syntax, lint, or compilation errors.
HTML Asset (dist/index.html)	0.77 kB (gzip: 0.44 kB)	Ultra-compact single-page host container.
CSS Bundle (dist/assets/*.css)	8.22 kB (gzip: 2.40 kB)	Minified Tailwind & Antigravity design tokens.
JS Bundle (dist/assets/*.js)	706.23 kB (gzip: 198.64 kB)	Includes React 18, Recharts SVG engine, and Axios.
Build Duration	7.37 Seconds (Clean compile)	Exceptional developer velocity and CI/CD readiness.

Summary Conclusion: The PredictStock frontend successfully transforms raw, complex machine learning outputs into a responsive, intuitive, and enterprise-grade operational decision-support tool. By leveraging React 18 concurrent features, Recharts telemetry, and glassmorphic styling tokens, the platform delivers high data density without compromising aesthetic elegance or operational usability.